

MA 5104: Statistical Inference and Learning

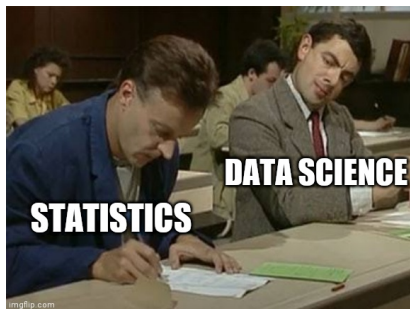
Module 1: Statistical Models and Likelihood

Dr. Rakhi Singh
Department of Mathematics
IIT Madras

Fall 2026

What Is This Course About?

- Why should we trust conclusions drawn from data?
- How do machines learn from data?
- What distinguishes explanation from prediction?
- How do we separate evidence from hype?



Source for the meme: web but unclear

Ex 1: Can Netflix Read Your Mind?

The Mystery

You watch one episode of *Stranger Things*.

The next day Netflix recommends

→ *Dark*

→ *Black Mirror*

→ *Wednesday*

but *not Friends*.

Ex 1: Can Netflix Read Your Mind?

The Mystery

You watch one episode of *Stranger Things*.

The next day Netflix recommends

- *Dark*
 - *Black Mirror*
 - *Wednesday*
- but *not Friends*.

How did Netflix know?



Ex 1: Can Netflix Read Your Mind? (contd.)

Available Data

- Viewing history
- Ratings and likes
- Search history
- Movie metadata
- Millions of users

Statistical Tools

- Statistical models
- Estimation
- Inference
- Prediction
- Model evaluation and Generalization

Ex 1: Can Netflix Read Your Mind? (contd.)

Available Data

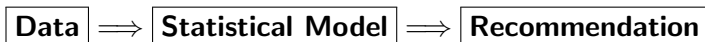
- Viewing history
- Ratings and likes
- Search history
- Movie metadata
- Millions of users

Statistical Tools

- Statistical models
- Estimation
- Inference
- Prediction
- Model evaluation and Generalization

Questions to answer

- Which movie should be recommended next?
- Will a user enjoy a particular movie?
- Which users have similar preferences?
- Which movies are similar?
- How should recommendation quality be evaluated?



Ex 2: Why Doesn't ChatGPT Give the Same Answer?

You and your friend ask ChatGPT

"Suggest an interesting PhD topic in Statistics."

The two answers are different.

Yet both are excellent.

Why?

Ex 2: Why Doesn't ChatGPT Give the Same Answer?

You and your friend ask ChatGPT

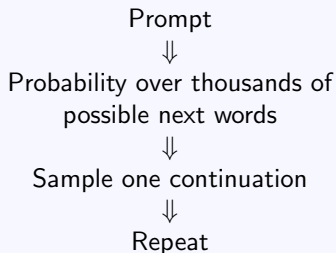
"Suggest an interesting PhD topic in Statistics."

The two answers are different.

Yet both are excellent.

Why?

One Possible Explanation



Ex 2: Why Doesn't ChatGPT Give the Same Answer?

You and your friend ask ChatGPT

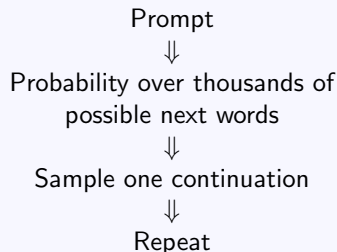
"Suggest an interesting PhD topic in Statistics."

The two answers are different.

Yet both are excellent.

Why?

One Possible Explanation



The statistical question

→ How can a machine learn an entire probability distribution over language rather than memorizing one “correct” answer?

Ex 3: Why Was My Credit Card Declined?

Yesterday

Coffee in Chennai; ₹180; Time: 11 AM

Today

Laptop in Prague; ₹2,40,000; Time:
2:13 AM

Card declined.

Why?

One Possible Explanation

Normal spending pattern



Today's transaction looks
unusual



Estimated fraud risk
= 98%



Decline

Ex 3: Why Was My Credit Card Declined?

Yesterday

Coffee in Chennai; ₹180; Time: 11 AM

Today

Laptop in Prague; ₹2,40,000; Time:
2:13 AM

Card declined.

Why?

One Possible Explanation

Normal spending pattern



Today's transaction looks
unusual



Estimated fraud risk
= 98%



Decline

The statistical question

→ How can a model learn what is “normal” for millions of customers and make reliable decisions in milliseconds?

Ex 4: Why Exactly 37 Minutes?

You check your flight.

Expected departure delay

37 minutes

Why exactly 37?

Ex 4: Why Exactly 37 Minutes?

You check your flight.

Expected departure delay

37 minutes

Why exactly 37?

One Possible Explanation

Late incoming aircraft

+22 min

Airport congestion

+11 min

Weather

+5 min

Early boarding

-1 min

Ex 4: Why Exactly 37 Minutes?

You check your flight.

Expected departure delay

37 minutes

Why exactly 37?

One Possible Explanation

Late incoming aircraft

+22 min

Airport congestion

+11 min

Weather

+5 min

Early boarding

-1 min

The statistical question

→ How can we combine many uncertain sources of information into one accurate prediction?

What Do These Examples Have in Common?

- They all begin with **data**, often large, noisy, and heterogeneous.
 - ▷ *Statistical Models and Likelihood*
- We seek to learn an underlying **statistical relationship**, rather than simply memorize past observations.
 - ▷ *Statistical Estimation*
- The learned model is used to make a **prediction**, estimate probabilities, rank alternatives, or support decisions.
 - ▷ *Prediction and Statistical Learning*
- Since predictions are imperfect, we must **quantify uncertainty** and evaluate how well the model generalizes to new data.
 - ▷ *Inference and Statistical Learning Theory*
- Throughout this course, we will develop the statistical principles underlying modern data-driven decision making.
 - ▷ *Statistical Thinking in Practice*

Beyond Learning Machine Learning

Question 1: What skills do you think companies look for today?

Beyond Learning Machine Learning

Question 1: What skills do you think companies look for today?

Question 2: If everyone has access to ChatGPT, what value can *you* add?

Beyond Learning Machine Learning

Question 1: What skills do you think companies look for today?

Question 2: If everyone has access to ChatGPT, what value can *you* add?

The answer

Companies do not just need people who can build models. They need people who can understand data, evaluate models, quantify uncertainty, and make trustworthy decisions.

The Skills Behind Modern AI

- Understand data and its limitations.
- Build **principled statistical** models.
- Estimate unknown quantities from data.
- Quantify uncertainty in predictions.
- Decide when a model can (or cannot) be trusted.
- Compare competing models fairly.
- Communicate evidence to support decision making.

These are statistical skills.

Learning Outcomes

By the end of the course, you should expect [‡] to

- Formulate statistical models for real-world data and perform **principled estimation and inference**.
- **Quantify uncertainty** using confidence intervals, hypothesis tests, and resampling methods.
- Develop and evaluate **predictive models** using modern statistical learning techniques.
- Assess model performance, understand generalization, and recognize the trade-offs between **model complexity and prediction accuracy**.
- **Critically evaluate quantitative claims** in scientific studies and the media, and communicate statistical findings effectively to diverse audiences.

[‡] granted that you work hard.

Module-wise distribution (1 of 2)

→ Module 1: Statistical Models and Likelihood

- ▷ How do we model uncertainty and explain observed data?
- ▷ How can machines uncover structure hidden in data?
- ▷ Statistical models, likelihood principle, maximum likelihood estimation, computational statistics.

→ Module 2: Statistical Estimation

- ▷ How can unknown quantities be estimated accurately?
- ▷ How can we learn from limited data without being misled by noise?
- ▷ Point estimation, properties of estimators, asymptotic theory, bootstrap methods.

→ Module 3: Statistical Inference

- ▷ How can we draw reliable conclusions from data?
- ▷ How can reliable decisions be made under uncertainty?
- ▷ Sampling distributions, confidence intervals, hypothesis testing, resampling-based inference.

Module-wise distribution (2 of 2)

→ Module 4: Prediction and Statistical Learning

- ▷ How can future observations be predicted?
- ▷ How do computers learn to recognize images, language, and patterns?
- ▷ Regression, classification, loss functions, predictive performance.

→ Module 5: Statistical Learning Theory and Generalization

- ▷ How can predictive models be evaluated and compared?
- ▷ Why do some AI models generalize while others overfit?
- ▷ Empirical risk minimization, cross-validation, regularization, model selection, ensemble learning.

→ Module 6: Statistical Thinking in Practice

- ▷ How should statistical evidence be communicated responsibly?
- ▷ How can we build trustworthy and reproducible AI systems?
- ▷ Reproducible analysis, statistical communication, case studies, student project presentations.

Grading

Your grade will be based on your performance on

- five **homeworks (10 percent)**, you cannot miss these; you will be given about a week to complete these, and these will tentatively be given at the end of each of the first five modules.
- **final project (15 percent)**, analyze a real-world statistical claim from a public dataset, scientific study, or news article (e.g., “Does drinking coffee increase life expectancy?”) by identifying bias, evaluating the evidence, and communicating the conclusions. Group size: 1-2; assessment: 5–10 minute presentation during the final week.
- **quizzes-1 and -2 (15 percent each)**, in institute-administered slots, no changes to these will be entertained.
- **endsem (45 percent)** in institute-administered slots, no changes to these will be entertained.

All of this is subject to Class Committee approval.

Structure of the class

We meet in **Slot B** so I will be generally teaching on

- **Mondays at 9**
- **Tuesdays at 8**
- **Wednesdays at 12**
- Fridays at 11 lecture will be utilized in case another lecture in the week is missed; will be communicated in advance

Throughout the class, I will show you some simulations where I may have used R/Python coding. HWs will also have some coding problems. Even if the coding problems will be ungraded, if you are doing this course for getting a job, it is advisable that you increase your familiarity with coding.

Prerequisites and Textbooks

Prerequisites.

I expect you to have the working knowledge of a rigorous first course in probability. I will not review probability concepts in this course, so please refresh them on your own before the next lecture. The course will be (and most certainly, will *feel like* it is) going fast-paced.

Textbooks.

- Efron, Bradley, and Trevor Hastie. Computer age statistical inference, student edition: algorithms, evidence, and data science. Vol. 6. Cambridge University Press, 2021.
- Hastie, Trevor, Tibshirani, Rob, and Friedman, Jerome, The Elements of Statistical Learning, 2nd ed., Springer, 2009.

You are not required to buy these books. You can essentially solve problems from any statistical inference book.

Use of Generative AI/LLM (GAI)

You may find it very tempting to just ask an LLM to solve the problem on your behalf. However, we need to ensure that GAI is not used as a substitute or replacement for student learning.

You may use them

- as a reference tool, similar to looking up the documentation for a function or Googling your problem.
- to assist in writing code in this class. You are expected to understand how any/all submitted code works.

You may not make use of the technology as a substitute for critical thinking. For example, you may not upload your data file to a GAI platform and ask it to create charts and statistical models for you.

You may not use GAI to write narratives for your assignments.

Statistician or data scientist perspective?

Fun (but true): <https://www.youtube.com/watch?v=uHG1Ci9j0WY>